

Spring 2026. Physics 510.

Quantum Theory II.

Homework # 5.

Due to April 6th, 2026.

Quantization of a Free Complex Scalar Field.

A free complex scalar field theory has a new fundamental feature, i.e., a $U(1)$ continuous symmetry and therefore an additional conservation law. It is a theory of a scalar doublet or complex scalar field $\phi = \phi_1 + i\phi_2$, with the global symmetry group $U(1)$. We are going to discuss the localization of this group and appearance of a vector field A_μ in our lectures soon.

Use my lectures about a real scalar field step by step as a pattern. Let's begin at the classical level.

Start with the action $S = \int \mathcal{L} d^4x$ where the Lagrange density for a free scalar field is given by the formula $\mathcal{L} = \frac{1}{2} \partial_\mu \phi^* \partial^\mu \phi - \frac{1}{2} m^2 \phi^* \phi$. Derive general Lagrange-Euler equations for $\mathcal{L} = \mathcal{L}(\phi, \partial^\mu \phi, x^\mu)$ using the least action principle. Make sure that for our particular case they yield the Klein-Gordon-Fock equation as we should expect.

Derive a formula for the energy of the field. Then use the $U(1)$ symmetry with respect to the global transformation $\phi \rightarrow e^{-i\alpha} \phi$ to derive a corresponding current $j_\mu = i(\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*)$ and an invariant charge $Q = \int j_0 d^3\vec{x}$. You can use either the Nöther theorem or the Klein-Gordon-Fock equation directly. Double-check that $\partial_\mu j^\mu = 0$ using the expression above and the Klein-Gordon-Fock equation.

Perform the Fourier transform $\phi(x) = \int e^{ikx} \tilde{\phi}(k) d^4k / (2\pi)^{3/2}$ and $\phi^*(x) = \int e^{ikx} \tilde{\phi}_*(k) d^4k / (2\pi)^{3/2}$. Note: $\tilde{\phi}_*(k) \neq \tilde{\phi}^*(k)$. Integrate over k^0 , and split the field into its positive and negative components $\phi^\pm(x) = \int \phi^\pm(\vec{k}) e^{\pm ikx} d^3\vec{k} / (2\pi)^{3/2} \sqrt{2k^0}$ and $\phi_*^\pm(x) = \int \phi_*^\pm(\vec{k}) e^{\pm ikx} d^3\vec{k} / (2\pi)^{3/2} \sqrt{2k^0}$. Again, $\phi_*^\pm(\vec{k}) \neq [\phi^\pm(\vec{k})]^*$ and $k^0 = \sqrt{\vec{k}^2 + m^2}$.

Plug in these expressions into the formulae for energy and charge and get the following expressions

$$E = \int k^0 \{ \phi_*^+(\vec{k}) \phi^-(\vec{k}) + \phi^+(\vec{k}) \phi_*^-(\vec{k}) \} d^3\vec{k}$$
$$Q = \int \{ \phi_*^+(\vec{k}) \phi^-(\vec{k}) - \phi^+(\vec{k}) \phi_*^-(\vec{k}) \} d^3\vec{k}$$

Now you see that $\phi_*^+(\vec{k})$ and $\phi^-(\vec{k})$ are classical analogs of creation and annihilation operators of a particle with a mass m , energy k^0 and an electric charge $+1$ while $\phi^+(\vec{k})$ and $\phi_*^-(\vec{k})$ are classical analogs of creation and annihilation operators of a particle with a mass m , energy k^0 and an electric charge -1 .

Switch to the quantum level and proceed with calculations following the real scalar field pattern. What kind of commutation relations should we expect?